

University of Dhaka

Department of Biomedical Physics & Technology

BMPT 514

Biomedical Instrumentation Laboratory

Experiment no. 02

Title

Student's name: Name

Session: 2024-2025

Group no. :01

Date of performance: 30 October 2019

Date of submission: 6 November 2019

Objective: The objective of this experiment is to

- get acquainted with the circuit diagram and data acquisition process in the ECG trainer kit
- and to observe trace of the six frontal plane ECG.

Theory:

The Electrocardiography (ECG) is the process of visualizing the electrical activity of the heart over a period of time. These electrical activities result from the depolarization of the myocardium in a certain pattern. The electrical activities are monitored using surface electrodes over the body.

As we know, the heart has a certain rhythm over a course of time in which it contracts and expands, pumping blood out to the body. This periodic rhythm is caused by depolarization of some specific points on the myocardium, namely the Sino-Atrial Node (SA node), the AtrioVentricular Node (AV node) and the His-Purkunje Fibers. These depolarizations create electrical impulses. As the human body is a volume conductor, the effect of any electrical impulse inside the body can be detected from the surface.

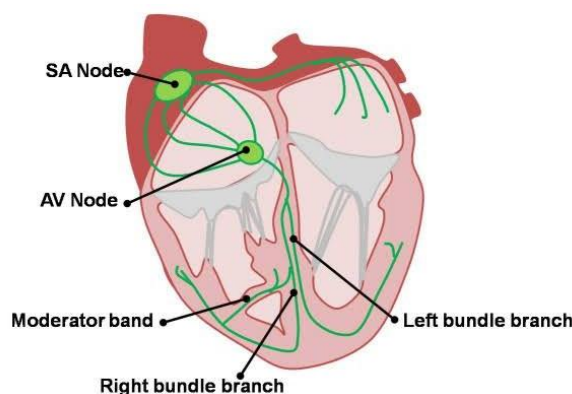


Figure: Nodes in the Myocardium

The electrical activity of the heart can be modelled as the movement of an electrical dipole. A dipole consists of two opposite charges. A cardiac vector is the line joining the two charges. Although the heart does not consist of two opposite charges, the electric field created due to the depolarization of the heart resembles that of an electric dipole. To describe a vector by its length in three dimensions, three directions are considered which are 60° apart. These directions form a triangle known as Einthoven's triangle. The direction from one point to another is known as a Lead, hence having three leads.

The body and the heart are three dimensional, so the electrical signals recorded from the skin will vary depending upon the position of the electrodes. For diagnostic purpose, some standardization of electrode positioning is needed. This is done by placing the electrodes on the limbs. The three points of the Einthoven's triangle are the three limbs where we put the electrodes. The three leads are thus considered as

Lead I: from Right Arm (-ve) to Left Arm (+ve)

Lead II: from Right Arm (-ve) to Left Leg (+ve)

Lead III: from Left Arm (-ve) to Left Leg (+ve)

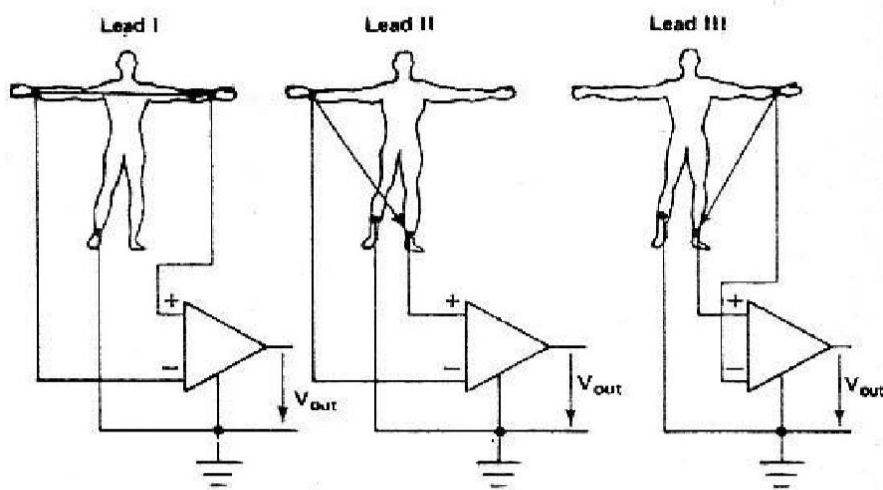


Figure: Limb leads

Apart from the three main leads, there are three other leads in the frontal plane named Augmented leads. These are denoted as aVR, aVL, and aVF. They are derived from the same three electrodes as leads I, II, and III, but they use Goldberger's central terminal as their negative pole. Goldberger's central terminal is a combination of inputs from two limb electrodes, with a different combination for each augmented lead. It is referred to immediately below as "the negative pole."

Lead aVR $RA - (LA + LL)/2$

Lead aVL $LA - (LL + RA)/2$

Lead aVF $LL - (RA + LA)/2$

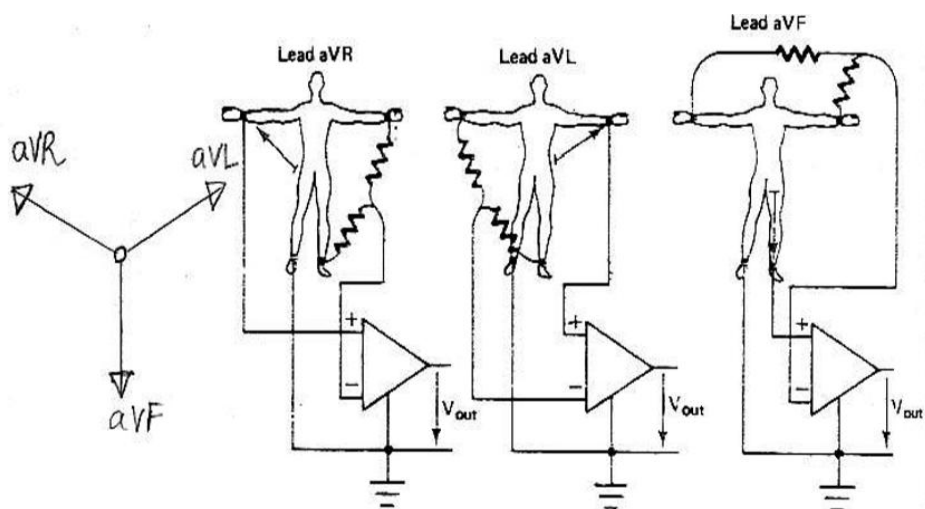


Figure: Augmented Virtual ECG leads

As the lower half of the body can be considered equipotential, the bottom point in the Einthoven's triangle can be considered as any point in the lower thorax. For simplicity, we use the left leg for this point.

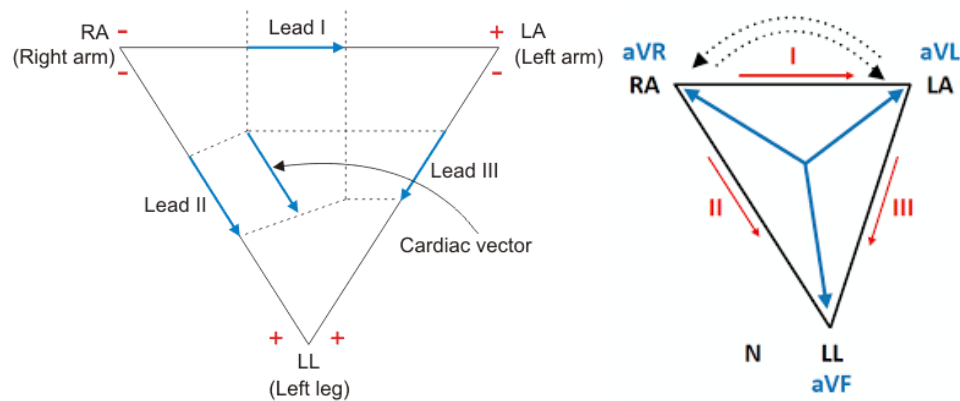
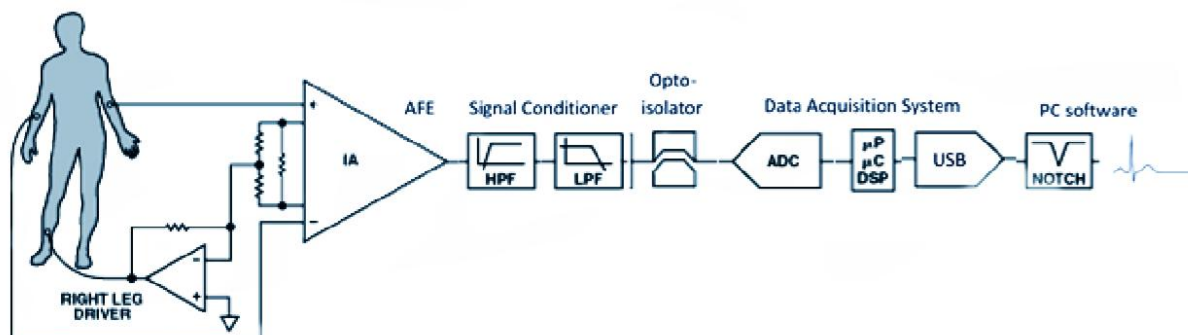


Figure: Einthoven Triangle showing the limb leads and augmented leads

To reduce the effect of noise, the average of the signals is fed back to the body via an instrumentation amplifier. This cancels out the common mode noise interference. As the left hand, right hand and left leg is used to record the various leads, this average signal is fed back to the body via the right leg. Hence this configuration is known as Right Leg driven circuit.

Module Diagram:



Equipment:

- ECG trainer module
- Computer / Oscilloscope

Data Table:

Horizontal scaling: 25mm/sec

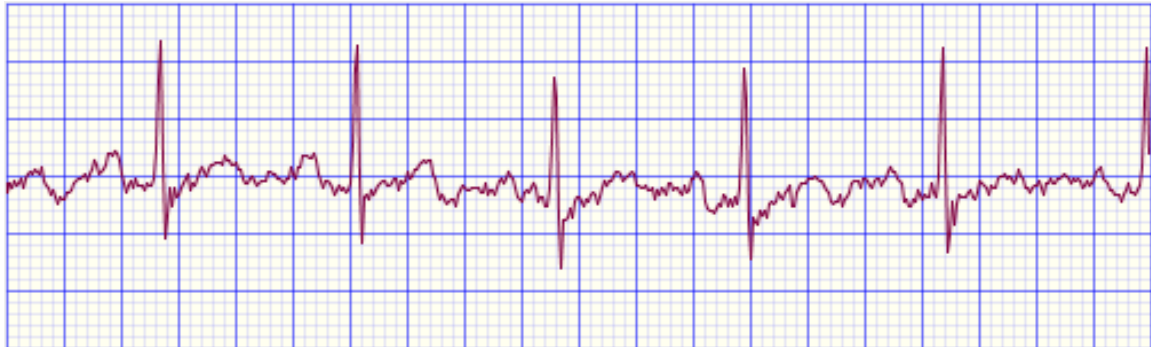
Vertical scaling: 10mm/mV

Time EG taken for is 4 second.

Specimen	No. of peaks	Duration (sec)	Heart beat rate (bpm)
1	6	4	90
2	8	4	120

Calculation:

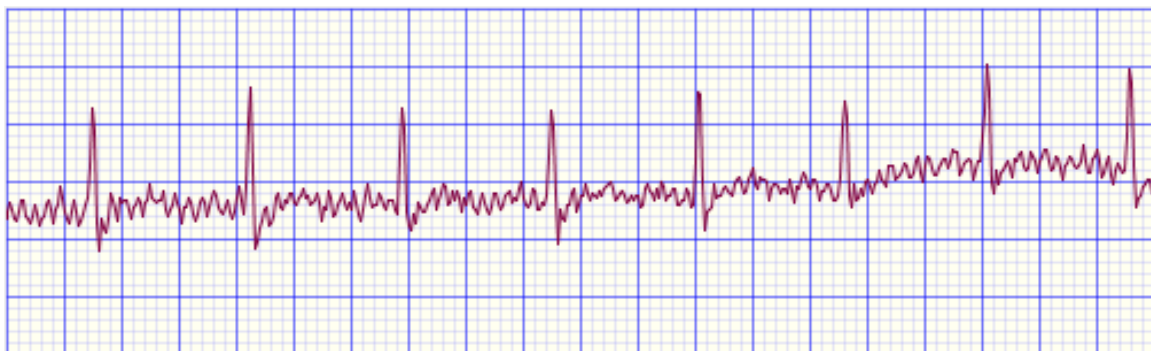
ECG plot of specimen 1 Lead II Gain 200 with respect to RLD: (amplitude vs time in second)



We have taken the ECG of 4 second.

So, heart beat rate of specimen 1 is $60 \times 6/4 = 90$ bpm.

ECG plot of specimen 2 Lead II Gain 200 with respect to RLD:

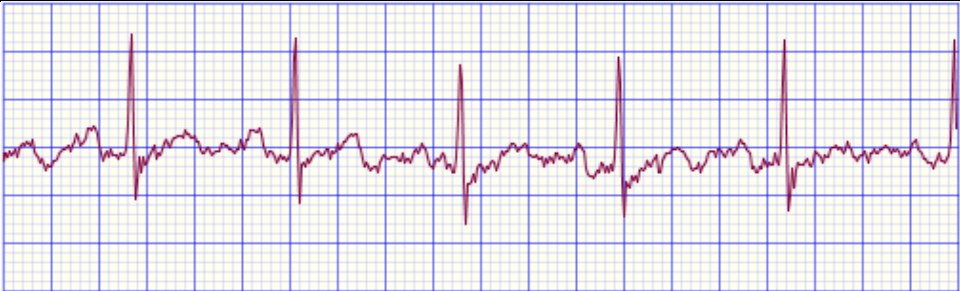
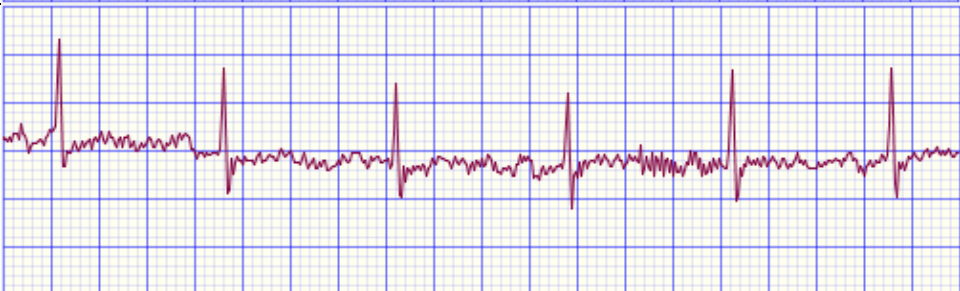
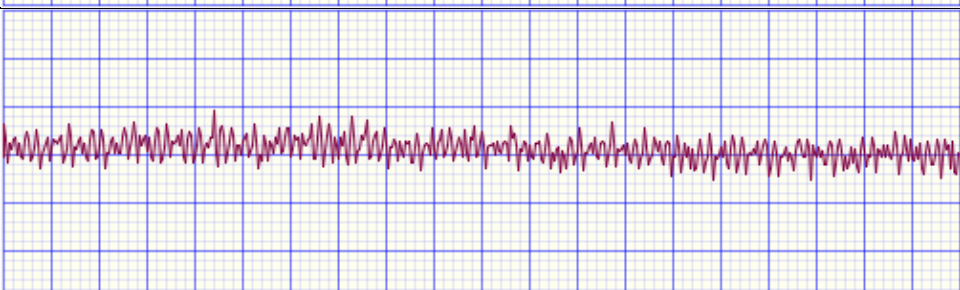
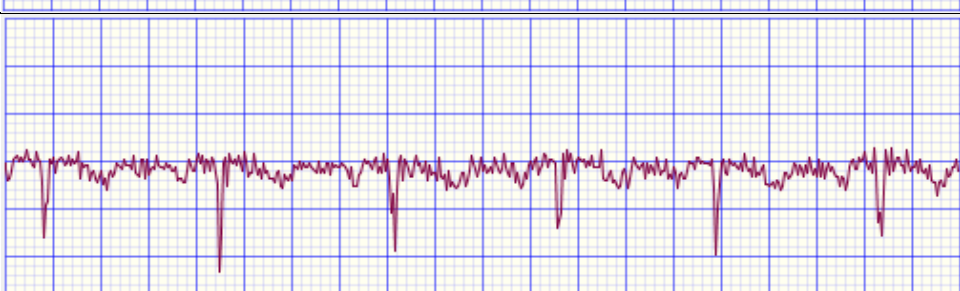
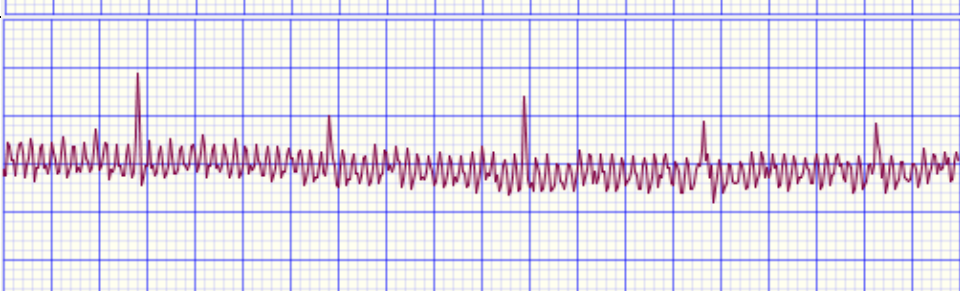


So, heart beat rate of specimen 2 is $60 \times 8/4 = 120$ bpm.

Graph:

ECG plot of specimen 1

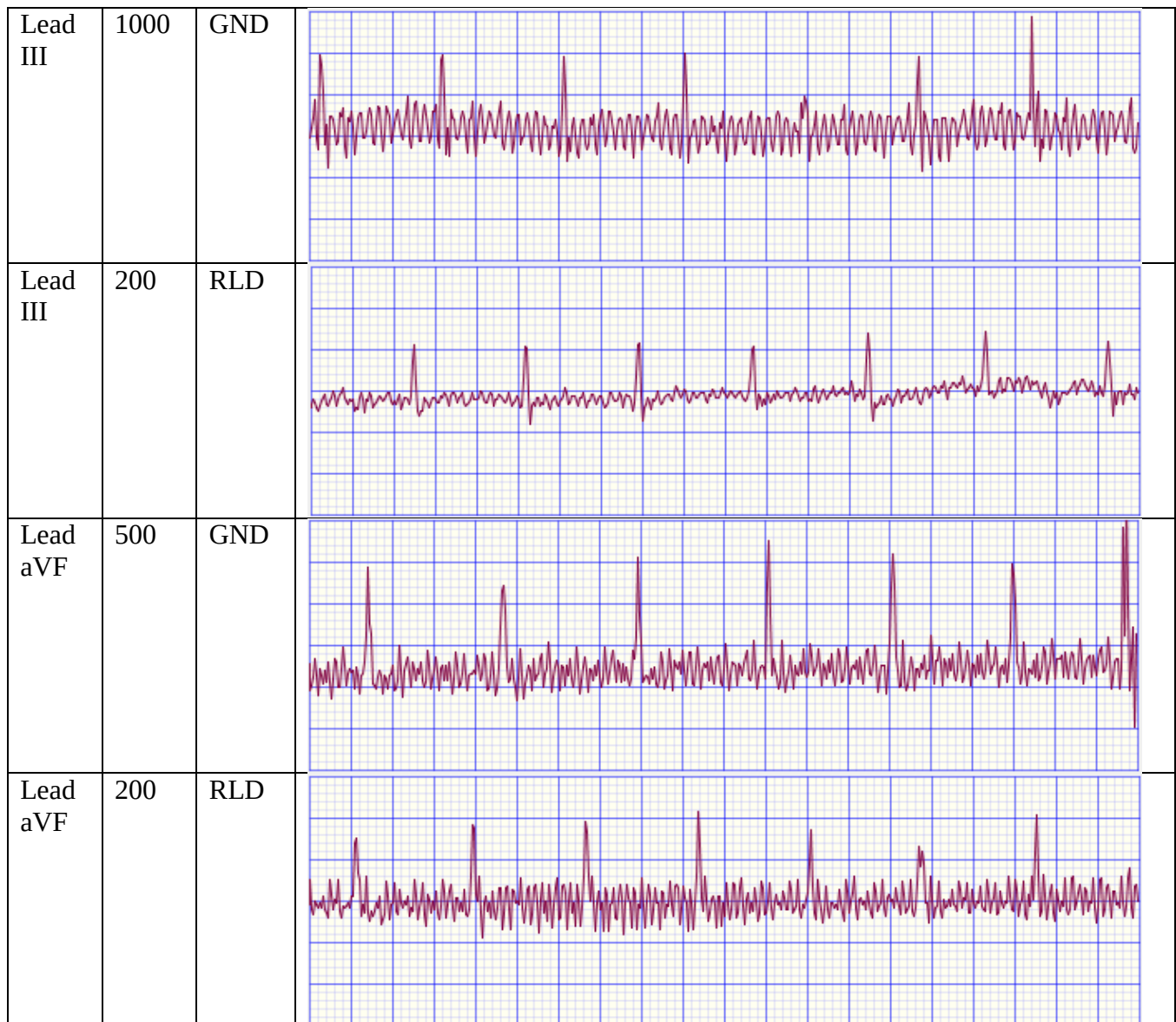
Lead	Gain	w.r.t	ECG Plot
Lead I	500	RLD	

Lead II	200	RLD	
Lead III	200	RLD	
Lead aVL	1000	RLD	
Lead aVR	1000	RLD	
Lead aVF	200	RLD	

ECG plot of specimen 2



Comparison of the reference in case of specimen 2:



Result:

Thus from this experiment, we got the plot of ECG of different leads for two specimens.

From the plots, first specimen heart beat rate: 90 bpm.

From the plots, second specimen heart beat rate: 120 bpm.

These values are approximate, not exact.

Discussion:

Using Frontal plane ECG trainer module we measured the heart beat rate. We have taken the ECG plot for all leads 4 second. From the graph we counted the peak points which denote the systole blood pressure. Using these data we calculated the heart beat rate in bpm. The data is taken in the analogue time domain. Circuitry systems are used to convert the continuous time signal into discrete time signal.

In horizontal scale the unit is 25mm/sec and in vertical scale it is 10mm/mV. As the signal is easily affected by artefacts, we need to be very careful while taking data. We avoided the transient period while receiving the ECG plot. Also there was a Digital filter which was kept on to avoid power line noise. The whole system needed a biasing voltage of 5V which was supplied from the source.

For specimen 1 we used only the electrodes to take data from the limbs. In this case a lot of noise affected the signal. But in case of specimen 2 we used the gel and hence a lot of noise was reduced.

The data is taken with respect to RLD in some cases and GND in 2 cases. It can be clearly seen that when the data is taken w.r.t RLD (Right Leg Driven) there is a huge decrease in noise than in case of GND. Actually the RLD is taken as the reference so that the noise generated from specimen body can be minimized using internal circuitry systems.

In case of plotting the ECG the gain plays an important role. Gain can be selected as 1000, 500, 200, 100. When the signal is at a very low amplitude or at a very high amplitude, the gain should be adjusted accordingly so that the plot is well enough estimable.

From the block diagram it can be seen that the analogue domain signal is first passed through a HPF so that if there is any DC offset or low frequency noise it can be removed easily. Then the LPF removes the high frequency noise including the power line noise.

There is an Opto-isolator which isolates the specimen from the electric circuit. This isolator ensures that the specimen is not connected with the actual ground of the circuit.

There is also a notch filter in the discrete time domain which removes the low frequency noise and high frequency noise in the discrete time domain.

In case of bipolar leads there is no null point. So the jumpers were directly connected to the respective points.

But in case of unipolar leads, there is a null point. So according to voltage divider rule, using two resistance of same value, the null point is detected. But there is a voltage drop in the resistant. So to solve this problem the other signal is also passed from a resistance of the same value. This resistance is named to be the augmented/chest lead. Using jumper, all the connections were made.

There is an inversion done internally by the circuit. So to solve this problem, the inverting input and non-inverting inputs were swapped. To avoid noise from artefact we tried to keep the whole thing free from vibration. To avoid transient part, we have taken the plot delaying some cycles.

There was a problem while calculating the heart rate of 2nd specimen. In some graph of 4 second duration, we got 7 peak points and in some cases we got 8 peak points. But as lead II has a very dominating and high value we have taken 8 peak points from the ECG of lead II.

Skin preparation is a very important step for getting the ECG plot. There remains a dead cell layer upon human skin which act as electrical isolation. So to remove the dead cell layer Hexasol or some kind of dead cell removing chemical need to be used.